Trovemo order lifecycle, end to end

As-built report of the **order** worktree assignment: placement, payment, CJ fulfilment, delivery and closure

Date 2026-10-10 | Worktree **order** (branch fix/order) | Modules litemall-order, litemall-core, litemall-db | Written by Claude Code for the Trovemo team

1. Summary

The assignment commissioned on 2026-09-05 (plan **plan-order-lifecycle-e2e.md**, findings F1 to F18, work packages A, B, C and the raises D) is **complete**. Every order-scoped finding is either fixed in code or closed by an explicit decision, all code is merged to master, and the order container running on trovemo.com since 2026-10-03/04 (master **8a71f78e7**) carries all of it. The litemall-order module test suite went from **298** to **400** tests, 0 failures, with no Flyway migration anywhere in the assignment.

What the work changed, in one paragraph: the store can no longer keep money for an order it has already cancelled; a CJ order that stalls, gets rejected, or gets cancelled at the supplier is now visible to the admin, reaches the ops mailbox, and in the cancelled case tells the customer; the customer is told when the order is delivered and can withdraw a refund request; a refund that cannot stop a CJ order that CJ has already paid for raises an alert instead of a silent timeline line; and production stopped logging every SQL statement with customer data.

What has still never happened: one real EUR order end to end in production. The plan names that as its actual acceptance. It is blocked on user-side items (CJ account balance, IOSS in the CJ dashboard), listed in section 9.

Milestone	Date	Commit	Tests (run / failures)	Deploy
Package A: payment money-safety (F1, F2)	2026-09-05	`c9ec31257`	342 / 0	2026-09-05 02:39 UTC, order container
Packages B + C: fulfilment visibility, closure	2026-09-05	`5cfe7b1d5`	387 / 0	2026-09-05 02:39 UTC, order container
Follow-ups: prod SQL DEBUG pin, F3 stock restore after commit	2026-09-13	`2cc0ba6a4`	395 / 0	2026-10-03/04 (inside 8a71f78e7)
Close-out: D5 ops alert, F13 documented	2026-09-20	`5bb240b23`, `81e805870`	400 / 0	2026-10-03/04 (inside 8a71f78e7)
gateway-api half (SPA, raised by this plan)	2026-09-13	`22a8d0390`	jest 57 suites / 392	2026-10-03/04 (inside 8a71f78e7)
gateway-admin half (parked rows + Requeue)	2026-09-23	`9fcd28726`	jest 100+	2026-10-03/04 (inside 8a71f78e7)

2. Why the work was needed

Verified on 2026-09-05 by reading the code and the dev database. The lifecycle existed in full on paper and had never run beyond "CJ order created" anywhere: dev had never produced a SHIPPED CJ order, prod had never placed one, and the mail outbox in prod was empty. The gaps clustered in four places:

- **Money could be taken for an order the store had already cancelled.** The unpaid sweep cancelled the local order after 30 minutes but never cancelled the Stripe PaymentIntent, and the intent id was not stored until payment succeeded. A late card confirmation, or any redirect method (Klarna, iDEAL, Bancontact, 3-DS) whose return never called the pay endpoint, charged the customer against a cancelled order and the webhook answered 200 with only an ERROR log.
- **After CJ placement the order went dark.** Local status stayed PAID until CJ said SHIPPED; a CJ order never paid from balance, a CJ cancellation, or a placement rejected after admin approval were log-only or invisible in the admin panel. The only recovery for a rejected order was a SQL statement printed into an ops email.

- **The customer was told almost nothing** between paid and shipped, was never told the order closed, and could not see the timeline the FAQ promised. A refund request (status 202) was a dead end with no actions.
- **Nothing had ever been exercised for real.**

Status codes used throughout: 101 CREATED, 102 CANCELED, 103 SYSTEM_CANCELED, 201 PAID, 202 REFUND_REQUEST, 203 REFUNDED, 301 SHIPPED, 401 DELIVERED, 402 AUTO_DELIVERED.

3. Decisions (approved 2026-09-05; D5 decided 2026-09-20)

#	Question	Decision
D1	Late payment lands on an order already cancelled	Automatic Stripe refund, timeline hop, customer mail. Never revive the order.
D2	CJ cancels an order the customer has paid	Automatic customer mail. No money moves automatically (the 2026-07-20 rule stands).
D3	Placement keeps failing with retryable errors	Warn ops after 60 min, park after 24 h under PLACEMENT_STALLED; admin requeues.
D4	Customer wants to take back a refund request	Yes: a customer withdraw action, 202 back to PAID or SHIPPED.
D5	Local refund after CJ already paid from balance	No automatic CJ dispute (a dispute is a product-problem claim with evidence; a store refund is not one). Ops alert instead.
D6	Who receives the operational alerts	Prod sets CJ_OPS_MAIL = contact@trovemo.com; a MAIN-session deploy step (done 2026-09-05).

4. Findings and their disposition

All eighteen findings from the plan, with where each one ended. "Order" means fixed in this worktree; the other owners are named.

#	Finding	Disposition
F1	Late payment on an auto-cancelled order kept, not refunded	Order, Package A: intent recorded at mint; sweep reconciles with Stripe first; stray charges refunded automatically.
F2	Transient verification failure burns the Stripe event id	Order, Package A: verify failure throws, webhook answers 503, claim rolled back, Stripe redelivers.
F3	Stock restored inside the cancel transaction, no rollback compensation	Order, follow-up 2026-09-13: restore moved to an afterCommit hook at both cancel sites.
F4	Local status never reflects fulfilment	Order, Package B: state lives on the timeline (hops + sentinels); SPA wording raised to gateway-api (built 22a8d0390).
F5	CJ order created but never paid from balance is silent forever	Order, Package B: cj_stall hop + ops mail at most every 24 h, facade returns CJ's reason.
F6	CJ CANCELLED after payment: no customer signal	Order, Package B (D2): fulfilment-cancelled customer mail + ops mail; no money moves.
F7	A parked order in manual mode is invisible and un-requeueable	Order, Package B: pending list includes parked rows with parkReason; POST .../cj-placement/requeue. Admin button: gateway-admin 9fcd28726.
F8	Retryable placement failures loop forever	Order, Package B (D3): one deferred hop, warn at 60 min, park at 24 h under PLACEMENT_STALLED.
F9	Open aftersale / refund request does not block approval or placement	Order, Package B: approve refuses [AFTERSALE_OPEN] / [PARKED]; place() holds on an open aftersale.
F10	Blank tracking number contradiction	Order, Package B: ship_sn stays NULL; /tracking answers TRACKING_PENDING for a shipped order.
F11	Sync predicate never retires 401/402	Order, Package B: predicate excludes 401/402; a CJ ship during a refund review leaves a hop.
F12	Timeline operator hard-coded 'admin' for the poller's ships	Order, Package B: the poller ships as operator 'system'.

#	Finding	Disposition
F13	Duplicate-placement guard is JVM-local	Closed as DOCUMENTED (2026-09-20): one order instance by compose design; sweep adopts an existing CJ order by order_sn; a durable claim would need V65.
F14	Delivered event has zero consumers; two auto-confirm sources of truth	Order, Package C: delivered mail; auto-confirm window = system setting litemall_order_unconfirm, yml fallback.
F15	Refund after CJ paid from balance performs no CJ-side action	Closed by D5 (2026-09-20): ops alert when a refund/cancel cannot stop a live CJ order; no automatic dispute.
F16	202 REFUND_REQUEST is a customer dead end	Order, Package C (D4): POST .../actions/refund/withdraw + handleOption.withdrawRefund. Refunds page: gateway-api 22a8d0390.
F17	Reviews have no purchase link	goods-management: purchase-gated reviews merged ebf75fb17 (2026-09-20), deploy pending at MAIN.
F18	Dead / contradictory helpers	Order, Package C: canBeCanceled delegates to the dispatcher, getStatusesForShowType deleted, dead branches and stale comments removed.

5. What was built: Package A, payment money-safety

Design record: litemall-order/docs/adr-stripe-payments.md section 11. Suite 298 to 342.

- **Intent recorded at mint.** createPaymentIntent persists the PaymentIntent id on the order when it is created (CREATED-guarded). A re-mint settles a succeeded previous intent, refuses while one is processing, and cancels a pending one at Stripe.
- **The unpaid sweep asks Stripe before cancelling.** UnpaidOrderTaskScheduler now owns only the task rows; UnpaidOrderReconciler decides: SUCCEEDED settles the order through settleVerifiedPspPayment; PROCESSING (SEPA in flight) defers 60 min; Stripe UNAVAILABLE defers 5 min; anything else cancels the intent AT STRIPE first, then the order. An order is never cancelled blind.
- **Stray charges are refunded.** A verified payment arriving (webhook or /actions/pay) for a cancelled or already-paid order is refunded under the idempotency key refund-order-<id>-<intent>, writes hop payment_refunded_stray and sends the customer the payment-refunded mail. A refund Stripe refuses writes payment_stray_unrefunded and mails ops.
- **Transient verify failures are retried by Stripe, not swallowed.** LitemallPaymentTemporarilyUnavailableException makes the webhook answer 503 with the event claim rolled back.
- Knobs: litemall.order.unpaid-reconcile.processing-defer-minutes and .unavailable-defer-minutes (env LITEMALL_ORDER_UNPAID_PROCESSING_DEFER_MINUTES / _UNAVAILABLE_DEFER_MINUTES).
- Load-bearing details: the orchestrator is injected @Lazy into the reconciler (a bean cycle); createPaymentIntent carries @Transactional(noRollbackFor = LitemallPaymentGatewayException) so the "already paid" refusal cannot roll back the settlement it just made; settleVerifiedPspPayment never touches the task table because the sweep holds FOR UPDATE SKIP LOCKED on it.

6. What was built: Package B, fulfilment visibility and control

Suite 342 to 387 together with Package C. The rule: nothing in CJ fulfilment fails silently any more, and the state is kept on the order timeline so no schema change was needed.

- **CjFulfilmentIncidentService** (@Transactional, because the after-commit mail listener drops events published outside a transaction) owns every "CJ went wrong" signal.
- **Retryable placement failures:** one "deferred" hop on the first failure; ops warning after litemall.order.cj.stall-warn-minutes (60); PARK under PLACEMENT_STALLED after litemall.order.cj.stall-park-hours (24).

- **Placed but CJ refuses to confirm or pay from balance** (empty CJ balance is the expected cause): cj_stall hop + ops mail at most every litemall.order.cj.lifecycle-alert-hours (24). The CJ facade returns CjCallOutcome carrying CJ's reason.
- **CJ CANCELLED after payment:** LitemallCjFulfilmentCancelledEvent, customer mail fulfilment-cancelled, ops mail. No money moves (D2).
- **Admin surface:** GET /srv/private/admin/order/cj-placement/pending now includes parked rows regardless of the approval stamp, with parked and parkReason (CJ's words from the last cj_placement_failed hop). POST /srv/private/admin/order/{id}/cj-placement/requeue clears either sentinel by compare-and-set and keeps the approval. Approve refuses [AFTERSALE_OPEN] and [PARKED].
- **Honest tracking:** a CJ ship with no tracking number keeps ship_sn NULL and /tracking answers TRACKING_PENDING. The sync predicate excludes 401/402. The poller ships as operator 'system'.

7. What was built: Package C, closure, plus the follow-ups and the D5 close-out

- **delivered mail** on LitemallOrderDeliveredEvent, stating the return window from litemall.order.return-window-days (30). Core MailTemplates and MailHtmlTemplates, same design shell as the other lifecycle mails.
- **Auto-confirm window** reads the system setting litemall_order_unconfirm when set (7 in the admin panel), yml fallback otherwise. Before, a hidden 15-day code default silently won.
- **Customer withdraw:** POST /srv/order/{id}/actions/refund/withdraw moves 202 back to PAID or SHIPPED; detail exposes handleOption.withdrawRefund.
- **Follow-up 1 (2026-09-13), prod SQL DEBUG firehose.** application.yml sets org.linlinjava.litemall.db to DEBUG and the prod yml countered only the parent package; Spring resolves the most specific key, so prod logged every MyBatis statement with bound customer data and Stripe ids, and the 30 MB json-file rotation evicted real errors within minutes. Pinned to INFO in application-prod.yml plus a compose passthrough LOGGING_LEVEL_ORG_LINLINJAVA_LITEMALL_DB (env LITEMALL_ORDER_DB_LOG_LEVEL). ProdLoggingConfigTest pins prod INFO and dev DEBUG. Prod proof after deploy: 0 SQL lines.
- **Follow-up 2 (2026-09-13), F3.** restoreStockOnCancelCommit replaces restoreStockForOrder at both cancel sites: the remote restore runs in an afterCommit synchronization, inline when no transaction is active. LitemallOrderCancelStockRestoreTest (6) is the module's first test class driving TransactionSynchronizationManager, including the never-on-rollback case.
- **Close-out (2026-09-20), D5.** CjFulfillmentService.cancelAtCjIfDeletable, the seam every refund-approve, aftersale-approve and cancel-during-placement path runs, now enqueues one ops mail through CjOpsNotifier in both not-cancelled branches: CJ status past CREATED/IN_CART/UNPAID (CJ already paid from balance), or CJ refused the delete. The outbox row is inserted in the caller's transaction, so a rolled-back refund takes its alert with it. Delete-succeeded, non-CJ and never-placed orders are unchanged. Blank CJ_OPS_MAIL stays log-only.

8. Operator reference

8.1 Configuration knobs (all env-backed with defaults; a container recreate applies an env change, no rebuild)

Property	Env var	Default	Meaning
`litemall.order.cj.placement-mode`	`LITEMALL_ORDER_CJ_PLACEMENT_MODE`	manual	Only the literal 'auto' disables the admin approval gate (Wave 23).
`spring.cjdropship.api.sandbox`	`LITEMALL_ORDER_CJ_SANDBOX`	true	Prod sets false. True would simulate CJ placement as a fake success.
`litemall.order.cj.stall-warn-minutes`	`LITEMALL_ORDER_CJ_STALL_WARN_MINUTES`	60	Ops warning for a placement still failing after this long.

Property	Env var	Default	Meaning
litemall.order.cj.stall-park-hours	LITEMALL_ORDER_CJ_STALL_PARK_HOURS	24	Park under PLACEMENT_STALLED after this long (D3).
litemall.order.cj.lifecycle-alert-hours	LITEMALL_ORDER_CJ_LIFECYCLE_ALERT_HOURS	24	Minimum gap between ops mails for a CJ order that will not confirm or pay.
litemall.order.unpaid-reconcile.processing-defer-minutes	LITEMALL_ORDER_UNPAID_PROCESSING_DEFER_MINUTES	60	Sweep defer while the Stripe intent is processing (SEPA).
litemall.order.unpaid-reconcile.unavailable-defer-minutes	LITEMALL_ORDER_UNPAID_UNAVAILABLE_DEFER_MINUTES	5	Sweep defer while Stripe is unreachable; never cancel blind.
litemall.order.return-window-days	(yml)	30	Return window named in the delivered mail.
litemall.order.auto-confirm-days	(yml)	fallback	Used only when litemall_system.litemall_order_unconfirm is unset.
(core) customer mail	LITEMALL_CUSTOMERMAIL_ENABLED, _USERNAME, _ADMIN_NOTIFY		Outbox swept every 60 s, max 5 attempts. USERNAME was missing in prod until 2026-10-03.
(core) ops alerts	CJ_OPS_MAIL	blank = log only	Prod: contact@trovemo.com since 2026-09-05.
logging	LITEMALL_ORDER_DB_LOG_LEVEL	INFO	MyBatis logger in prod. DEBUG prints bound customer data.

8.2 Endpoints added or changed

Method and path	Who	Behaviour
GET /srv/private/admin/order/cj-placement/pending	admin	Paid CJ orders awaiting approval AND parked rows (parked, parkReason, holdReason, cjReady).
POST /srv/private/admin/order/{id}/cj-placement/approve	admin	Stamps approval (idempotent). Typed 422 refusals: [NOT_FOUND] [NOT_CJ] [NOT_PAID] [ALREADY_PLACED] [AFTERSALE_OPEN] [PARKED].
POST /srv/private/admin/order/{id}/cj-placement/requeue	admin	CAS-clears PLACEMENT_REJECTED or PLACEMENT_STALLED, keeps the approval; the sweep retries on its next tick.
POST /srv/order/{id}/actions/refund/withdraw	customer	202 back to PAID or SHIPPED while no refund has settled (D4).
GET /srv/order/{id}/timeline	customer	Pre-existing; now rendered by the storefront (gateway-api 22a8d0390).
GET /srv/order/{id}/tracking	customer	Answers TRACKING_PENDING for a shipped order with no number yet.
POST /srv/order/stripe/webhook	Stripe	Stray charge refunded; transient verify failure answers 503 so Stripe redelivers.

8.3 Customer and ops mails (litemall-core templates, HTML + plain-text twins, outbox-backed)

Template key	Trigger	Recipient
order-confirmation	Order paid (Wave 10)	customer
payment-refunded	Stray payment refunded (D1, Package A)	customer
fulfilment-cancelled	CJ cancelled a paid order (D2, Package B)	customer
shipped	CJ or admin ships (pre-existing)	customer
delivered	Delivered or auto-confirmed, names the return window (Package C)	customer
refund-approved, pickup-code	Pre-existing lifecycle mails	customer
admin paid-order notice	Order paid (Wave 23)	LITEMALL_CUSTOMERMAIL_ADMIN_NOTIFY
CJ ops alerts	Stall warn, park, lifecycle failure, CJ cancelled, refund-with-live-CJ-order (D5), stray refund refused	CJ_OPS_MAIL

8.4 Timeline sentinels and hops to recognise

Marker	Where	Means
`PLACEMENT_REJECTED`	cj_order_status	CJ refused the placement after approval (e.g. 7001 missing IOSS). Visible in pending with parkReason; Requeue after fixing the cause.
`PLACEMENT_STALLED`	cj_order_status	Retryable failures for more than stall-park-hours. Requeue restarts the attempts.
`cj_stall`	timeline hop	Placed at CJ but CJ will not confirm or pay from balance; ops mailed.
`cj_placement_failed / deferred`	timeline hop	Last CJ error text; the first failure writes exactly one deferred hop.
`payment_refunded_stray`	timeline hop	A payment for a non-payable order was refunded at Stripe (D1).
`payment_stray_unrefunded`	timeline hop	Stripe refused the refund; ops mailed; needs a manual refund.
`TRACKING_PENDING`	/tracking status	Shipped at CJ, tracking number not yet supplied.

9. Production state and what remains open

9.1 Deploy record

- **2026-09-05 02:39 UTC:** packages A, B, C deployed; only the order image rebuilt and recreated; CJ_OPS_MAIL=contact@trovemo.com appended to .env.prod (backup .env.prod.bak-lifecycle-20260905). Boot verified: Stripe ENABLED (currency=eur), CJ ENABLED (sandbox=false), placement mode MANUAL, Flyway validated 64. The nested litemall-core jar was verified to contain the three new mail templates.
- **2026-10-03/04:** order, gateway-api and gateway-admin recreated from master 8a71f78e7, images tagged build-8a71f78e7, rollback tags pre-20261003. Jars were built on the workstation, not on the VPS (a compose build OOM-killed the host earlier that day). This deploy carries 2cc0ba6a4 and the D5 close-out: order SQL DEBUG lines in prod are 0. The same session found and fixed the missing LITEMALL_CUSTOMERMAIL_USERNAME, which had made every mail send fail with Brevo 502 "Please authenticate first".
- Nothing in this assignment needed a migration; prod schema is unchanged by it (V60 at the time of the first deploy, Flyway 64/65 validated at boot later).

9.2 Prod facts met by the new code (as of the 2026-10-03 read-only audit)

Order	State	What to do
7, 10	Paid 2026-07/08 with the pre-live Stripe TEST keys; intents do not exist on the live account; unapproved	Never approve for CJ. Cancel locally; no Stripe refund is possible for a test-mode intent.
9	202 refund request open since 2026-07-23, also a test-mode payment	Cannot be refunded through live Stripe; settle administratively.
11	Approved 2026-08-09, rejected by CJ 7001 'Please enter a IOSS number', PLACEMENT_REJECTED. The live F7 case, now visible with parkReason	Set IOSS in the CJ dashboard, then Requeue from the admin pending tab.
all	No CJ order has ever been placed in prod; the mail outbox had never held a row before 2026-10-03	The first real order is the acceptance of the whole plan.

9.3 Open items, with owners

Item	Owner	Why it matters
Top up the CJ account balance	user	Auto-pay-from-balance: an approved order sits at CJ UNPAID with balance 0 (the exact F5 case the stall alert now reports).
Set IOSS in the CJ dashboard	user	Every EU order is rejected 7001 until then; order 11 waits on it.
One real small EUR order end to end	user	Pay, approve, CJ pay-from-balance, ship, tracking, delivered mail. The plan's acceptance; never done.
Deploy goods-management (F17 purchase-gated reviews, Wave 28 origin)	MAIN	Merged ebf75fb17 on 2026-09-20, not yet in a prod image.

Item	Owner	Why it matters
Standalone /pay/:orderId page mounts no Stripe Elements	gateway-api	Every unpaid order reached from the order list is a dead end (pre-existing since Wave 7).
Tokenized guest order view	gateway-api + order	Mail CTAs send Wave-16 guest buyers to a login wall. Order side would mint a signed, expiring token into the confirmation mail.
F13 durable placement claim	nobody	Only if the order service is ever scaled past one instance (needs V65).

10. Lessons recorded for the next person

- **A sweep that keeps rows "for retry" is safe only if every permanent failure is non-throwing at its source**; otherwise it is an infinite loop (the 35-day cancelled-order loop of 2026-08-27 and the F8 placement loop share this shape).
- **Spring resolves the most specific logger key.** A prod yml that counters org.linlinjava.litemall at INFO loses to application.yml's org.linlinjava.litemall.db DEBUG. Counter at the same specificity and pin it with a resource test.
- **Remote side effects belong after the commit.** A Feign call that commits in another service (stock restore, pink release, coupon release) must run in an afterCommit synchronization, inline only when no transaction is active.
- **Record the PaymentIntent id at mint, not at success,** or there is nothing to cancel or reconcile when the order times out. Ask the PSP before cancelling an order it may already have charged.
- **Events published outside a transaction are dropped by an AFTER_COMMIT listener.** Services that publish lifecycle mails must be @Transactional (the shipped-mail lesson, applied to CjFulfilmentIncidentService).
- **Alerts that ride the caller's transaction cannot cry wolf:** inserting the outbox row inside the refund transaction means a rolled-back refund takes its alert with it.
- **"Deployed" is proven by the running artifact, not by the checkout.** Verify the jar inside the container (class names in the zip listing, nested litemall-core bytes) and verify env inside the container.
- **Worktree merges to master move the ref with git push . fix/order:master** when the MAIN checkout holds uncommitted edits; never touch MAIN's tree from a worktree, and never build on the production VPS.
- Test harness gotchas: Mockito here has no verifyNoInteractions; argThat lambdas in a second when() see null; a Java record component named ok forbids a static factory ok(); LitemallCjRetryableException wraps its message (match with contains).

11. References (all in the repository)

Document	What it holds
`litemall-order/docs/plan-order-lifecycle-e2e.md`	The contract: findings F1-F18, packages A/B/C, raises D, decisions D1-D6, acceptance.
`litemall-order/docs/plan-order-followups-2026-09-06.md`	Prod SQL DEBUG pin and F3 stock restore after commit (built 2026-09-13).
`litemall-order/docs/plan-order-closeout-2026-09-20.md`	D5 decision and ops alert, F13 kept documented, raises found on the way.
`litemall-order/docs/adr-stripe-payments.md section 11`	Design of the Package A payment money-safety changes.
`litemall-order/docs/handoff-gateway-api-lifecycle.md`	Contract the storefront coded to (payment copy, redirect return, withdraw, timeline).
`litemall-order/docs/handoff-gateway-admin-cj-requeue.md`	Contract the admin coded to (parked rows, Requeue, approve refusals).
`docker-compose/order-flow-check.sh`	Read-only prod order-flow check (OK / WARN / FAIL), from the 2026-10-03 audit.

Document	What it holds
`CLAUDE.md, block 'Worktree: order'`	Running status log of this assignment, including deploy evidence.